Electric Vehicle Population Data — Complete EDA This notebook performs a comprehensive Exploratory Data Analysis on EV population data, featuring dark-themed visualizations and dataset splitting based on Electric Range.

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')

plt.rcParams.update({
    'figure.facecolor': '#0F1117',
    'axes.facecolor':   '#1A1D27',
    'axes.edgecolor':   '#3A3D4A',
    'axes.labelcolor':  '#E0E0E0',
    'xtick.color':      '#B0B0B0',
    'ytick.color':      '#B0B0B0',
    'text.color':       '#E0E0E0',
    'grid.color':       '#2A2D3A',
    'grid.linestyle':   '--',
    'grid.alpha':       0.5,
    'font.family':      'DejaVu Sans',
    'axes.titlesize':   10,
    'axes.labelsize':   8,
    'xtick.labelsize':  7,
    'ytick.labelsize':  7,
})

# Dataset color palette:  Blue = All | Red = Range > 0 | Green = Range = 0
PALETTE = ['#4F86F7', '#F75A4F', '#4FF77B']
ALPHA   = 0.80

print("Libraries imported successfully.")
```

```
Libraries imported successfully.
```

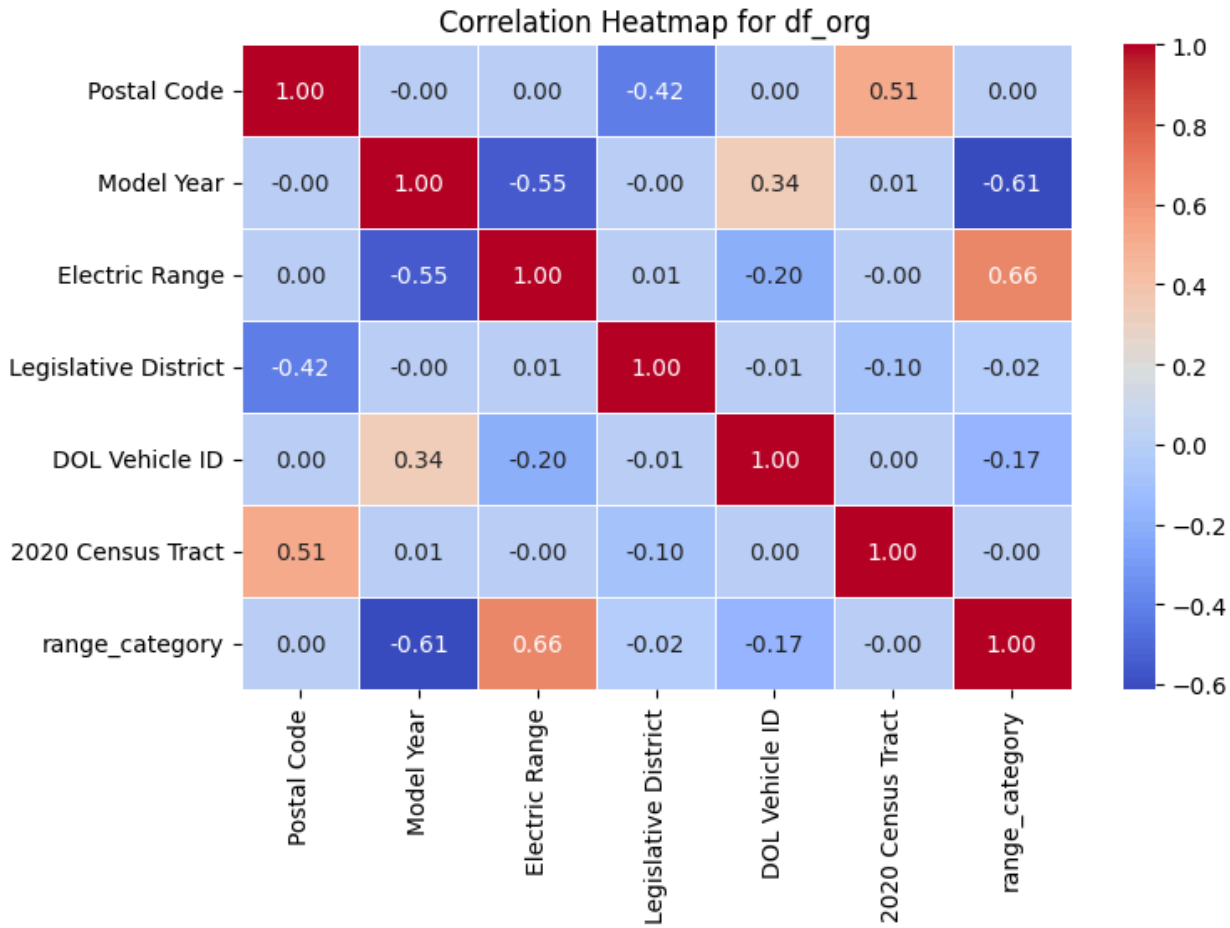
In [2]:
```
# Load the dataset
df = pd.read_csv('Electric_Vehicle_Population_Data.csv')

# Create splits based on range
df_pos  = df[df['Electric Range'] > 0].copy()
df_zero = df[df['Electric Range'] == 0].copy()

datasets = [df,    df_pos,       df_zero    ]
labels   = ['All', 'Range > 0',  'Range = 0']
colors   = PALETTE

print(f"All     rows : {len(df):,}")
print(f"Range > 0   : {len(df_pos):,}  ({len(df_pos)/len(df)*100:.1f}%)")
print(f"Range = 0   : {len(df_zero):,}  ({len(df_zero)/len(df)*100:.1f}%)")
print("\nColumns:", df.columns.tolist())
```

```
All     rows : 279,780
Range > 0   : 101,832  (36.4%)
Range = 0   : 177,937  (63.6%)

Columns: ['VIN (1-10)', 'County', 'City', 'State', 'Postal Code', 'Model Year',
'Make', 'Model', 'Electric Vehicle Type', 'Clean Alternative Fuel Vehicle (CAF
V) Eligibility', 'Electric Range', 'Legislative District', 'DOL Vehicle ID', 'V
ehicle Location', 'Electric Utility', '2020 Census Tract']
```

Plotting Heatmap of correlation. Legislative district and cencus tract has no contribution to Electric Range

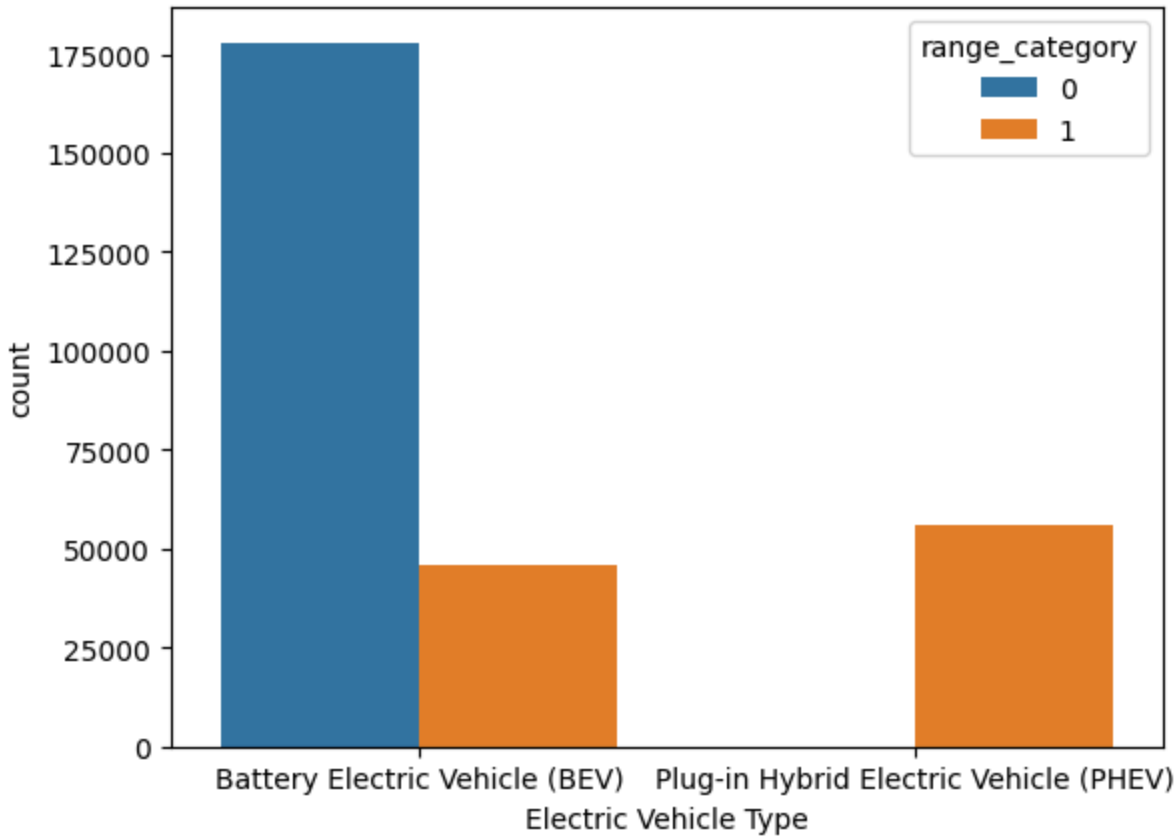
In [16]:

```
# 1. Calculate the correlation matrix
# Note: numeric_only=True ensures we only calculate correlation for numbers
corr_matrix = df_org.corr(numeric_only=True)

# 2. Set up the matplotlib figure size
plt.figure(figsize=(8, 5))

# 3. Create the heatmap
sns.heatmap(corr_matrix,
            annot=True,       # Shows the numbers in the cells
            cmap='coolwarm',  # Color scheme (red for positive, blue for negat
            fmt=".2f",        # Rounds numbers to 2 decimal places
            linewidths=0.5)   # Adds thin lines between cells for clarity

# 4. Add a title and show the plot
plt.title('Correlation Heatmap for df_org')
plt.show()
```



In [9]:
```
df_org['range_category'] = np.where(df_org['Electric Range'] > 0, 1, 0)
sns.countplot(x='Electric Vehicle Type',hue='range_category',data=df_org)
```

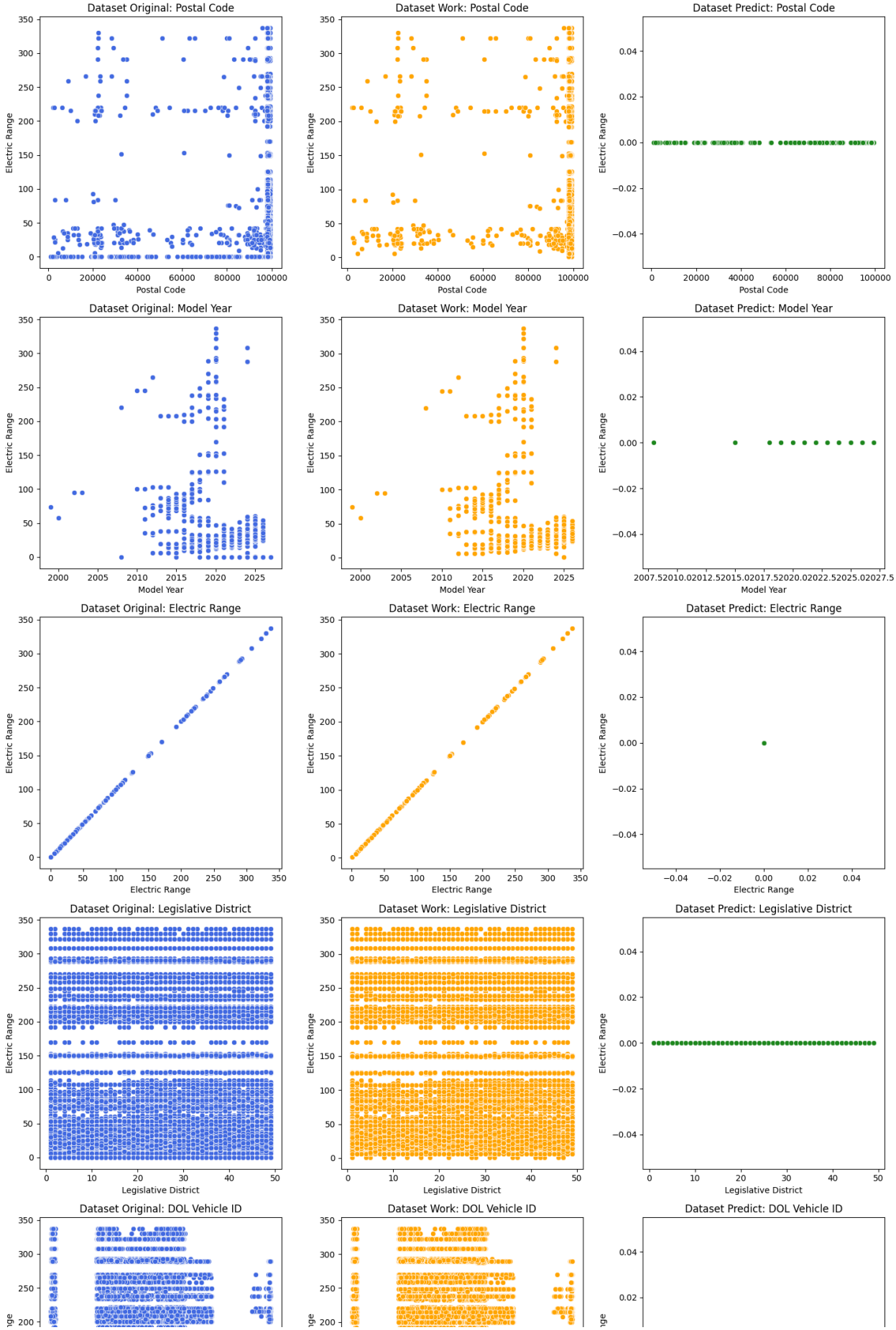
Out[9]: <Axes: xlabel='Electric Vehicle Type', ylabel='count'>



Relation between model year and Electric Range

In [15]:
```
# 1. Adjusted to 6 rows and 3 columns to match your 'axes' indexing
fig, axes = plt.subplots(nrows=6, ncols=3, figsize=(15, 30))

# 3. Iterate through the columns
for i, col in enumerate(cols_num):
    # Ensure we don't go out of bounds if cols_num has more than 6 items
    if i >= 6: break

    # FIX: x=col (the variable), not x='col' (the string literal)

    # Plot Dataset Original
    sns.scatterplot(x=col, y='Electric Range', ax=axes[i, 0], color='royalblue
    axes[i, 0].set_title(f'Dataset Original: {col}', fontsize=12)

    # Plot Dataset Work
    sns.scatterplot(x=col, y='Electric Range', ax=axes[i, 1], color='orange',
    axes[i, 1].set_title(f'Dataset Work: {col}', fontsize=12)

    # Plot Dataset Predict
    # FIX: Corrected index to axes[i, 2] for the third column
    sns.scatterplot(x=col, y='Electric Range', ax=axes[i, 2], color='forestgre
    axes[i, 2].set_title(f'Dataset Predict: {col}', fontsize=12)

# 4. Clean up the labels and layout
plt.tight_layout()
```

```
plt.show()
```


Dataset Original: Postal Code
Dataset Work: Postal Code
Dataset Predict: Postal Code
Electric Range
Postal Code
Dataset Original: Model Year
Dataset Work: Model Year
Dataset Predict: Model Year
Model Year
Dataset Original: Electric Range
Dataset Work: Electric Range
Dataset Predict: Electric Range
Dataset Original: Legislative District
Dataset Work: Legislative District
Dataset Predict: Legislative District
Legislative District
Dataset Original: DOL Vehicle ID
Dataset Work: DOL Vehicle ID
Dataset Predict: DOL Vehicle ID

In [3]:
```
print("=" * 55)
print("SHAPE & DTYPES")
print("=" * 55)
print(df.dtypes)

print("\n" + "=" * 55)
print("MISSING VALUES")
print("=" * 55)
print(df.isnull().sum())

print("\n" + "=" * 55)
print("ELECTRIC RANGE — DESCRIPTIVE STATS")
print("=" * 55)
print(df['Electric Range'].describe())
```

```
==========================================================
SHAPE & DTYPES
==========================================================
VIN (1-10)                                            object
County                                                object
City                                                  object
State                                                 object
Postal Code                                          float64
Model Year                                             int64
Make                                                  object
Model                                                 object
Electric Vehicle Type                                 object
Clean Alternative Fuel Vehicle (CAFV) Eligibility     object
Electric Range                                       float64
Legislative District                                 float64
DOL Vehicle ID                                         int64
Vehicle Location                                      object
Electric Utility                                      object
2020 Census Tract                                    float64
dtype: object

==========================================================
MISSING VALUES
==========================================================
VIN (1-10)                                             0
County                                                24
City                                                  24
State                                                  0
Postal Code                                           24
Model Year                                             0
Make                                                   0
Model                                                  0
Electric Vehicle Type                                  0
Clean Alternative Fuel Vehicle (CAFV) Eligibility      0
Electric Range                                        11
Legislative District                                 700
DOL Vehicle ID                                         0
Vehicle Location                                     109
Electric Utility                                      24
2020 Census Tract                                     24
dtype: int64

==========================================================
ELECTRIC RANGE — DESCRIPTIVE STATS
==========================================================
count    279769.000000
mean         39.172256
std          78.230356
min           0.000000
25%           0.000000
50%           0.000000
75%          32.000000
max         337.000000
Name: Electric Range, dtype: float64
```

In [4]:
```
def style_ax(ax, title='', xlabel='', ylabel=''):
    """Apply consistent dark-theme styling to an axes."""
    ax.set_title(title, fontsize=10, fontweight='bold', pad=6)
    ax.set_xlabel(xlabel, fontsize=8)
    ax.set_ylabel(ylabel, fontsize=8)
    ax.tick_params(axis='x', rotation=30)
    ax.grid(True, axis='y', alpha=0.4)
    ax.spines[['top', 'right']].set_visible(False)

def add_badge(ax, label, color):
    """Add a colored dataset badge inside the top-right corner of an axes."""
    ax.text(
        0.98, 0.97, label,
        transform=ax.transAxes,
        ha='right', va='top',
        fontsize=8, fontweight='bold', color='white',
        bbox=dict(boxstyle='round,pad=0.3', facecolor=color, alpha=0.85, edgec
    )

print("Helper functions ready.")
```

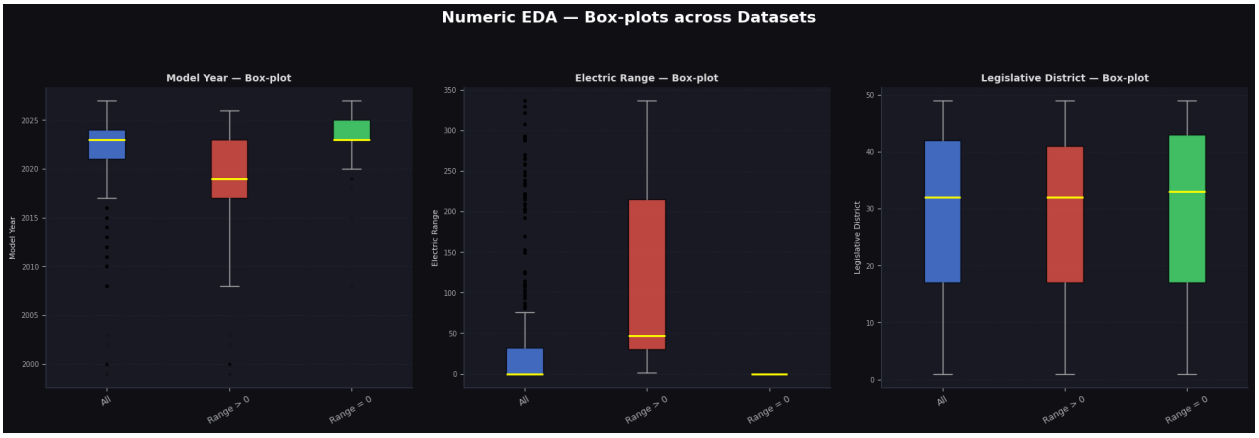
```
Helper functions ready.
```

Section 1: Numeric EDA Exploring distributions, outliers, and statistical comparisons across numeric features.

In [5]:
```
num_cols_plot = ['Model Year', 'Electric Range', 'Legislative District', 'Post

fig, axes = plt.subplots(
    len(num_cols_plot), 3,
    figsize=(18, 5 * len(num_cols_plot)),
    facecolor='#0F1117'
)
fig.suptitle('Numeric EDA — Histograms (All / Range > 0 / Range = 0)',
             fontsize=16, fontweight='bold', color='white', y=1.01)

for row, col in enumerate(num_cols_plot):
    for c_idx, (dset, lbl, clr) in enumerate(zip(datasets, labels, colors)):
        ax   = axes[row, c_idx]
        data = dset[col].dropna()

        ax.hist(data, bins=40, color=clr, alpha=ALPHA, edgecolor='none')
        style_ax(ax, title=col, xlabel=col, ylabel='Count')
        add_badge(ax, lbl, clr)

        ax.axvline(data.mean(),   color='white',  lw=1.2, linestyle='--', labe
        ax.axvline(data.median(), color='yellow', lw=1.2, linestyle=':',  labe

        if row == 0 and c_idx == 0:
            ax.legend(fontsize=7, facecolor='#1A1D27', edgecolor='none', label

plt.tight_layout(pad=2.0)
plt.show()
```

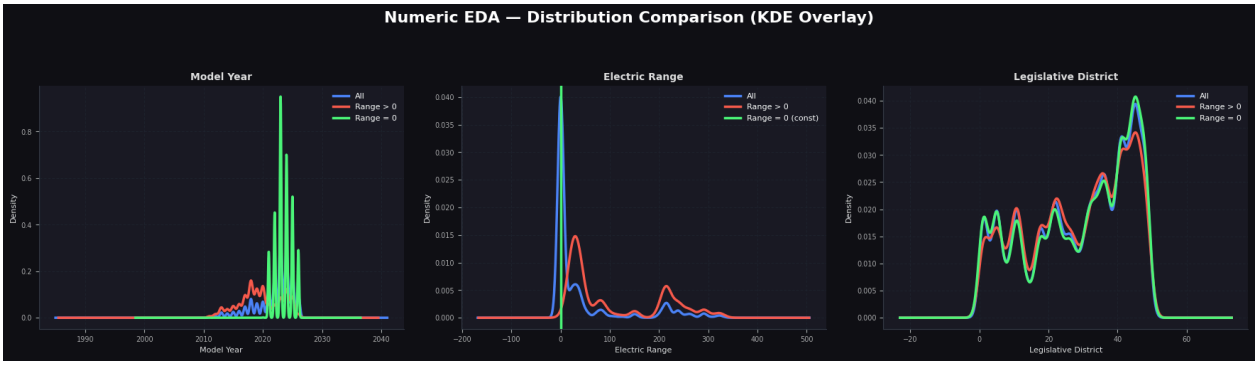
In [6]:
```
num_cols = ['Model Year', 'Electric Range', 'Legislative District']

fig, axes = plt.subplots(1, 3, figsize=(18, 6), facecolor='#0F1117')
fig.suptitle('Numeric EDA — Box-plots across Datasets',
             fontsize=16, fontweight='bold', color='white', y=1.02)

for ax, col in zip(axes, num_cols):
    box_data = [dset[col].dropna() for dset in datasets]

    bp = ax.boxplot(
```

```
        box_data,
        patch_artist=True,
        notch=False,
        medianprops=dict(color='yellow', linewidth=2),
        whiskerprops=dict(color='#B0B0B0'),
        capprops=dict(color='#B0B0B0'),
        flierprops=dict(marker='o', color='#B0B0B0', alpha=0.2, markersize=2)
    )
    for patch, clr in zip(bp['boxes'], colors):
        patch.set_facecolor(clr)
        patch.set_alpha(0.75)

    ax.set_xticks([1, 2, 3])
    ax.set_xticklabels(labels, fontsize=9)
    style_ax(ax, title=f'{col} — Box-plot', ylabel=col)

plt.tight_layout(pad=2.0)
plt.show()
```

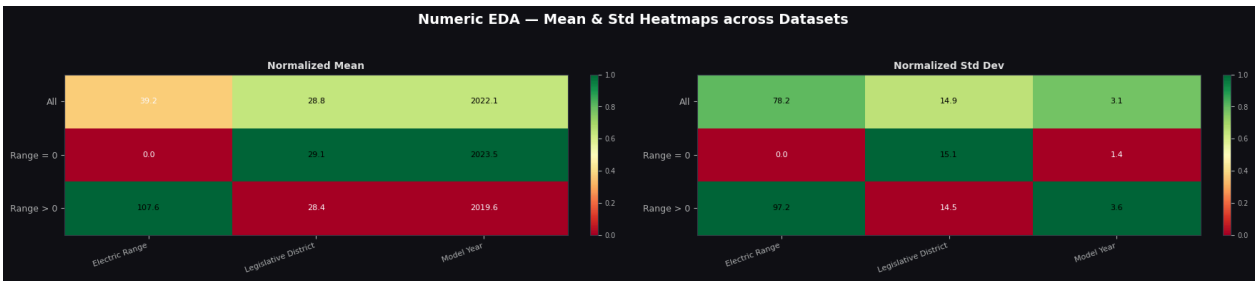


In [7]:
```
fig, axes = plt.subplots(1, 3, figsize=(18, 5), facecolor='#0F1117')
fig.suptitle('Numeric EDA — Distribution Comparison (KDE Overlay)',
             fontsize=16, fontweight='bold', color='white', y=1.02)

for ax, col in zip(axes, num_cols):
    for dset, lbl, clr in zip(datasets, labels, colors):
        data = dset[col].dropna()
        if data.nunique() <= 1:
            ax.axvline(data.iloc[0] if len(data) else 0,
                       color=clr, linewidth=2.5, label=f'{lbl} (const)')
        else:
            data.plot.kde(ax=ax, label=lbl, color=clr, linewidth=2.5)

    ax.set_title(col, fontweight='bold')
    ax.set_xlabel(col)
    ax.set_ylabel('Density')
    ax.legend(fontsize=8, facecolor='#1A1D27', edgecolor='none', labelcolor='w
    ax.spines[['top', 'right']].set_visible(False)
    ax.grid(True, alpha=0.4)

plt.tight_layout(pad=2.0)
```

```
plt.show()
```

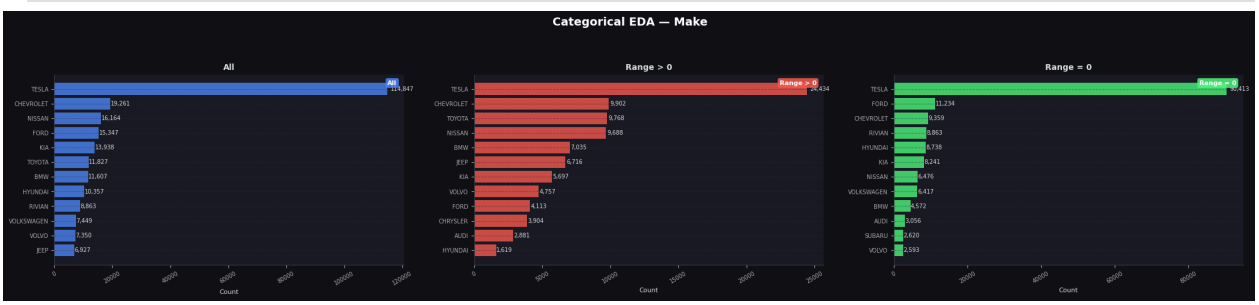


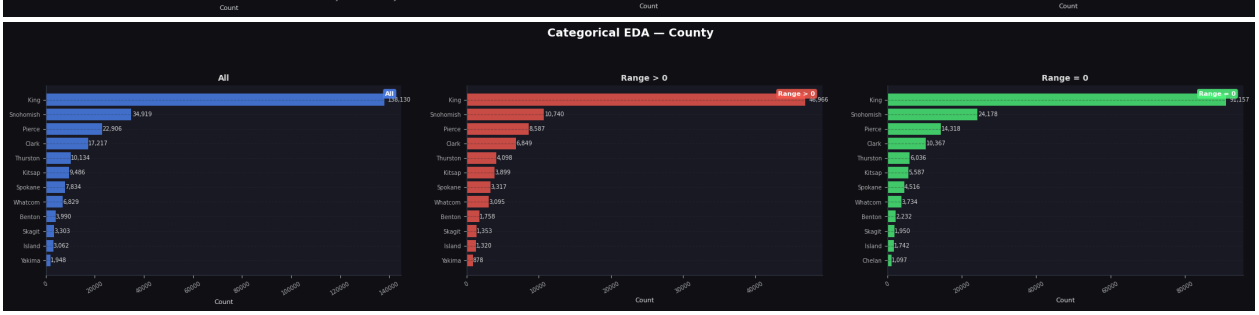
In [8]:
```
stat_rows = []
for dset, lbl in zip(datasets, labels):
    stats = dset[num_cols].describe().T
    stats.insert(0, 'Dataset', lbl)
    stat_rows.append(stats)

stats_df   = pd.concat(stat_rows)
pivot_mean = stats_df.pivot_table(index='Dataset', values='mean',
                                  columns=stats_df.index.get_level_values(0))
pivot_std  = stats_df.pivot_table(index='Dataset', values='std',
                                  columns=stats_df.index.get_level_values(0))

fig, axes = plt.subplots(1, 2, figsize=(18, 4), facecolor='#0F1117')
fig.suptitle('Numeric EDA — Mean & Std Heatmaps across Datasets',
             fontsize=14, fontweight='bold', color='white')

for ax, piv, title_s in zip(axes, [pivot_mean, pivot_std], ['Mean', 'Std Dev']
    norm_piv = piv.apply(lambda x: (x - x.min()) / (x.max() - x.min() + 1e-9))
    im = ax.imshow(norm_piv.values, cmap='RdYlGn', aspect='auto', vmin=0, vmax

    ax.set_xticks(range(len(norm_piv.columns)))
    ax.set_xticklabels(norm_piv.columns, rotation=20, ha='right', fontsize=8)
    ax.set_yticks(range(len(norm_piv.index)))
    ax.set_yticklabels(norm_piv.index, fontsize=9)
    ax.set_title(f'Normalized {title_s}', fontweight='bold')

    for i in range(len(norm_piv.index)):
        for j in range(len(norm_piv.columns)):
            val = piv.values[i, j]
            ax.text(j, i, f'{val:.1f}', ha='center', va='center',
                    fontsize=8,
                    color='black' if norm_piv.values[i, j] > 0.5 else 'white')

    plt.colorbar(im, ax=ax, fraction=0.046, pad=0.04)

plt.tight_layout(pad=2.0)
plt.show()
```

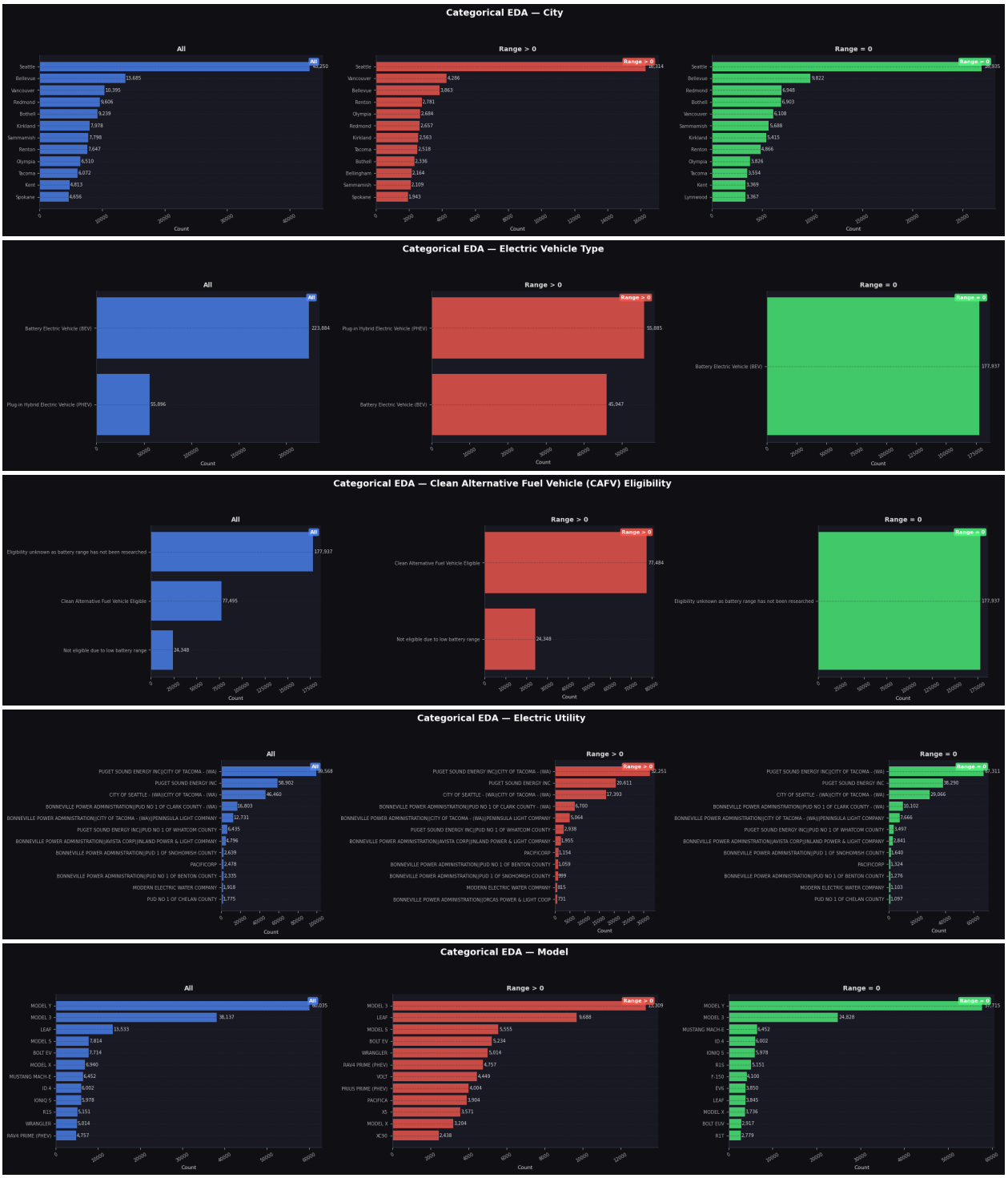


Section 2: Categorical EDA Analyzing frequencies and shares for Makes, Counties, EV Types, and more.

In [9]:
```
TOP_N = 12
categories = ['Make', 'County', 'City', 'Electric Vehicle Type',
              'Clean Alternative Fuel Vehicle (CAFV) Eligibility', 'Electric U

for cat in categories:
    fig, axes = plt.subplots(1, 3, figsize=(22, 5), facecolor='#0F1117')
    fig.suptitle(f'Categorical EDA — {cat}', fontsize=14, fontweight='bold', c

    for ax, dset, lbl, clr in zip(axes, datasets, labels, colors):
        counts = dset[cat].value_counts().head(TOP_N)
        bars   = ax.barh(counts.index[::-1], counts.values[::-1],
                         color=clr, alpha=ALPHA, edgecolor='none')
        for bar, val in zip(bars, counts.values[::-1]):
            ax.text(bar.get_width() * 1.01, bar.get_y() + bar.get_height() / 2
                    f'{val:,}', va='center', fontsize=7, color='#E0E0E0')
        style_ax(ax, title=lbl, xlabel='Count', ylabel='')
        add_badge(ax, lbl, clr)
        ax.tick_params(axis='y', labelsize=7)

    plt.tight_layout(pad=2.0)
    plt.show()
```

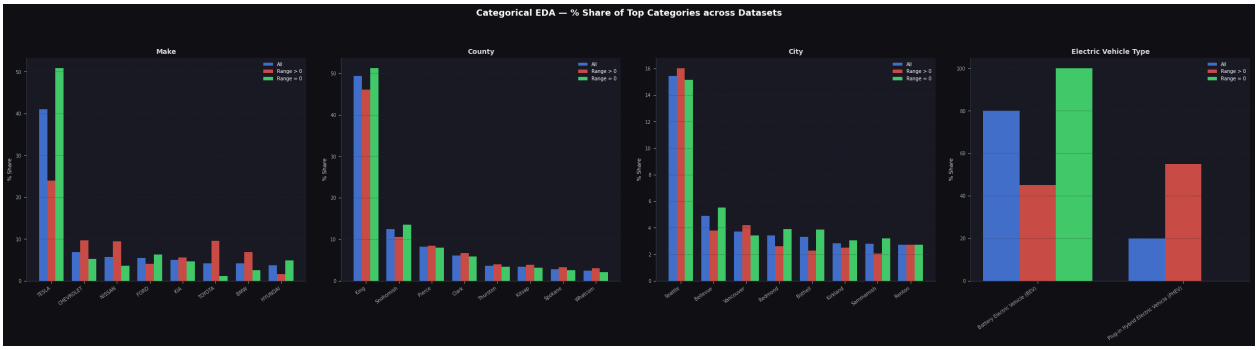




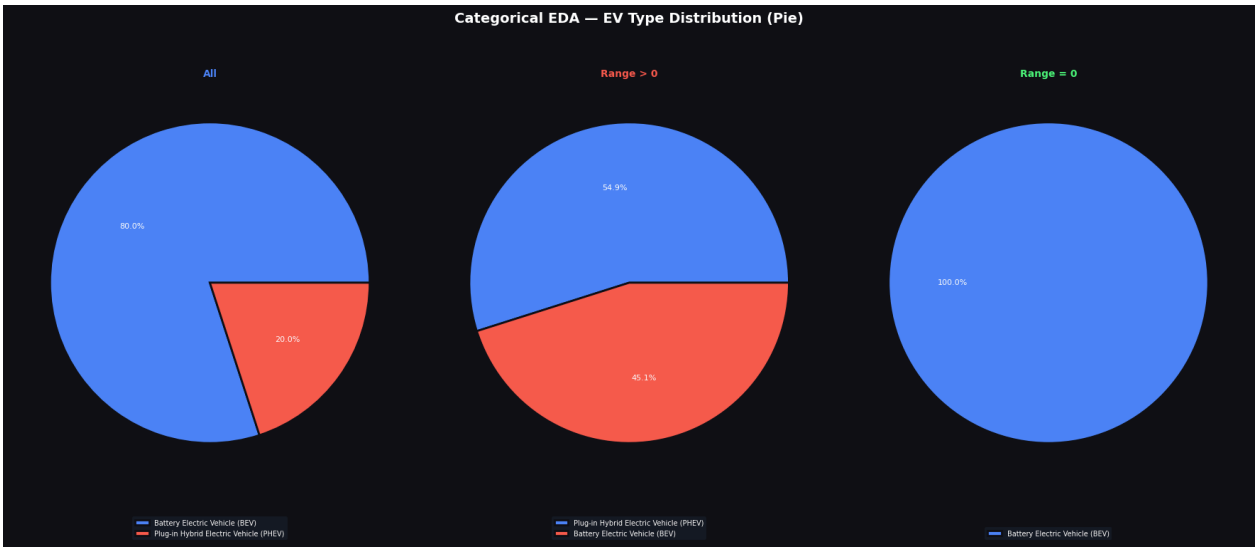


In [10]:
```
compare_cats = ['Make', 'County', 'City', 'Electric Vehicle Type']

fig, axes = plt.subplots(1, 4, figsize=(26, 7), facecolor='#0F1117')
fig.suptitle('Categorical EDA — % Share of Top Categories across Datasets',
             fontsize=13, fontweight='bold', color='white', y=1.01)

for ax, cat in zip(axes, compare_cats):
    top_cats = df[cat].value_counts().head(8).index
    x        = np.arange(len(top_cats))
```

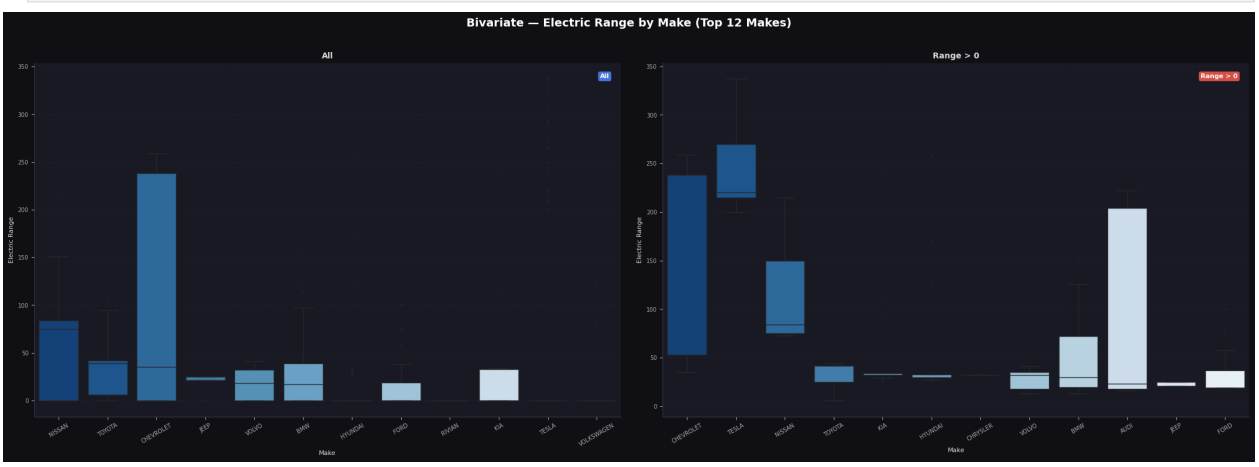
```
    width    = 0.25

    for i, (dset, lbl, clr) in enumerate(zip(datasets, labels, colors)):
        counts = dset[cat].value_counts()
        pcts   = [(counts.get(c, 0) / len(dset) * 100) for c in top_cats]
        ax.bar(x + i * width, pcts, width,
               label=lbl, color=clr, alpha=ALPHA, edgecolor='none')

    ax.set_xticks(x + width)
    ax.set_xticklabels(top_cats, rotation=35, ha='right', fontsize=7)
    ax.set_title(cat, fontweight='bold')
    ax.set_ylabel('% Share')
    ax.legend(fontsize=7, facecolor='#1A1D27', edgecolor='none', labelcolor='w
    ax.spines[['top', 'right']].set_visible(False)
    ax.grid(True, axis='y', alpha=0.4)

plt.tight_layout(pad=2.0)
plt.show()
```



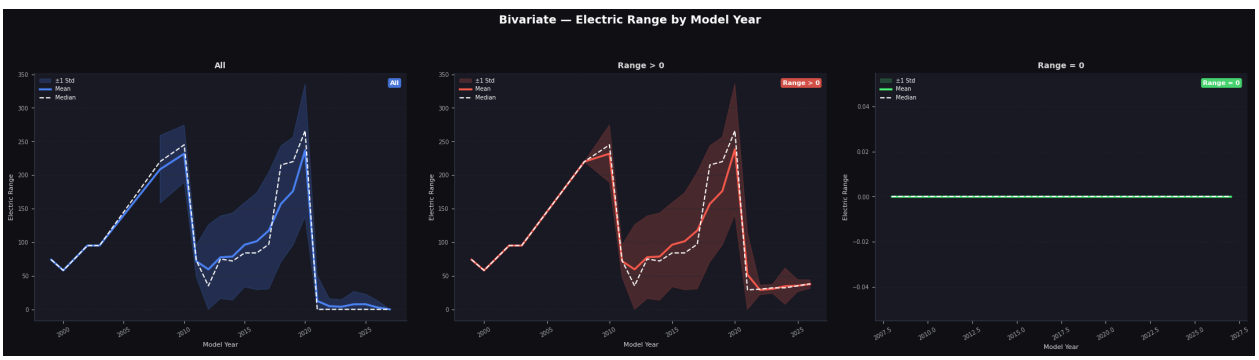
In [11]:
```
pie_colors = ['#4F86F7', '#F75A4F', '#4FF77B', '#F7C94F']

fig, axes = plt.subplots(1, 3, figsize=(18, 7), facecolor='#0F1117')
fig.suptitle('Categorical EDA — EV Type Distribution (Pie)',
             fontsize=14, fontweight='bold', color='white', y=1.01)

for ax, dset, lbl, ring_clr in zip(axes, datasets, labels, colors):
    vc = dset['Electric Vehicle Type'].value_counts()
    wedges, texts, autotexts = ax.pie(
        vc.values,
        labels=None,
        autopct='%1.1f%%',
        colors=pie_colors[:len(vc)],
        wedgeprops=dict(edgecolor='#0F1117', linewidth=2),
        textprops=dict(color='white', fontsize=8)
    )
    ax.legend(wedges, vc.index,
              loc='lower center', fontsize=7,
              facecolor='#1A1D27', edgecolor='none', labelcolor='white',
              bbox_to_anchor=(0.5, -0.15))
    ax.set_title(lbl, fontweight='bold', color=ring_clr)

plt.tight_layout(pad=2.0)
plt.show()
```



In [12]:
```
fig, axes = plt.subplots(1, 2, figsize=(22, 8), facecolor='#0F1117')
fig.suptitle('Bivariate — Electric Range by Make (Top 12 Makes)',
             fontsize=14, fontweight='bold', color='white')

for ax, (dset, lbl, clr) in zip(axes, [(df, 'All', colors[0]),
                                        (df_pos, 'Range > 0', colors[1])]):
    top_makes = dset['Make'].value_counts().head(12).index
    sub       = dset[dset['Make'].isin(top_makes)]
    order     = (sub.groupby('Make')['Electric Range']
                    .median()
                    .sort_values(ascending=False)
                    .index)
    sns.boxplot(
        data=sub, x='Make', y='Electric Range', order=order,
        palette=sns.color_palette('Blues_r', 12), ax=ax,
        flierprops=dict(marker='o', alpha=0.2, markersize=2)
    )
    style_ax(ax, title=lbl, xlabel='Make', ylabel='Electric Range')
    add_badge(ax, lbl, clr)

plt.tight_layout(pad=2.0)
plt.show()
```

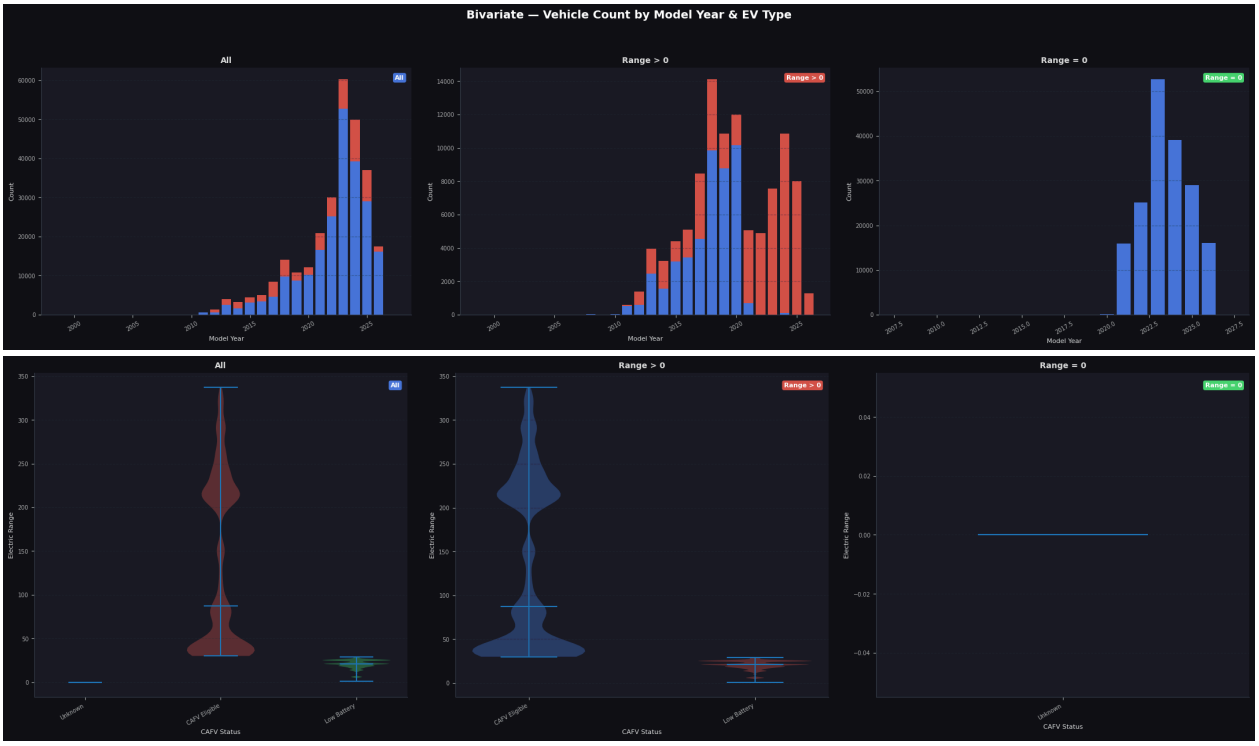


In [13]:
```
fig, axes = plt.subplots(1, 3, figsize=(22, 6), facecolor='#0F1117')
fig.suptitle('Bivariate — Electric Range by Model Year',
             fontsize=14, fontweight='bold', color='white', y=1.01)

for ax, dset, lbl, clr in zip(axes, datasets, labels, colors):
    grp = (dset.groupby('Model Year')['Electric Range']
               .agg(['mean', 'median', 'std'])
               .reset_index())

    ax.fill_between(
        grp['Model Year'],
        (grp['mean'] - grp['std']).clip(lower=0),
        grp['mean'] + grp['std'],
        color=clr, alpha=0.2, label='±1 Std'
    )
    ax.plot(grp['Model Year'], grp['mean'],   color=clr,    linewidth=2.5, lab
    ax.plot(grp['Model Year'], grp['median'], color='white', linewidth=1.5,
            linestyle='--', label='Median')

    style_ax(ax, title=lbl, xlabel='Model Year', ylabel='Electric Range')
    add_badge(ax, lbl, clr)
    ax.legend(fontsize=7, facecolor='#1A1D27', edgecolor='none', labelcolor='w

plt.tight_layout(pad=2.0)
plt.show()
```

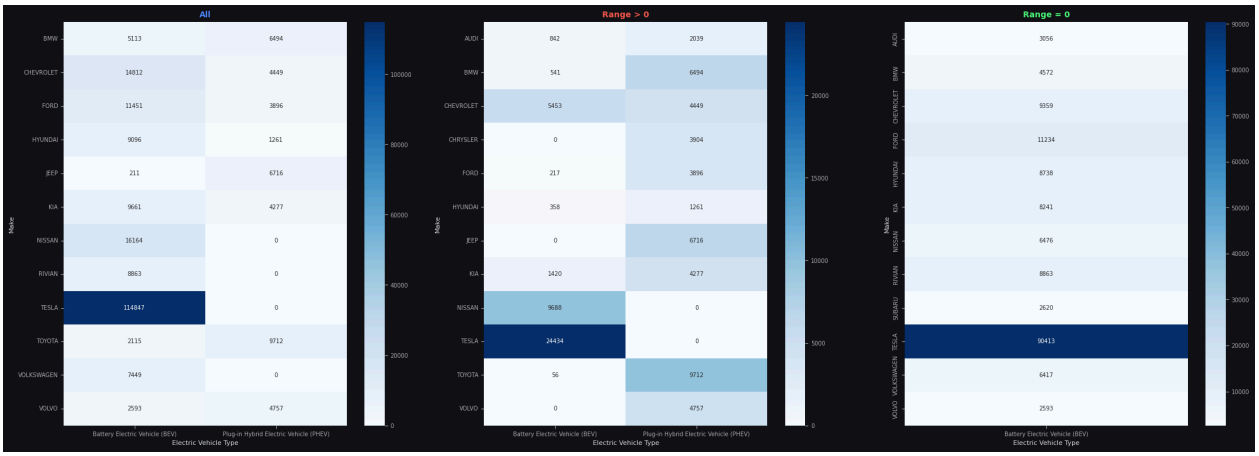


In [14]:
```
# Stacked Bar: Count by Year & Type
ev_colors = ['#4F86F7', '#F75A4F', '#4FF77B', '#F7C94F']
fig, axes = plt.subplots(1, 3, figsize=(22, 6), facecolor='#0F1117')
fig.suptitle('Bivariate — Vehicle Count by Model Year & EV Type', fontsize=14,

for ax, dset, lbl, clr in zip(axes, datasets, labels, colors):
    pivot = (dset.groupby(['Model Year', 'Electric Vehicle Type']).size().unst
    years, bottoms = pivot.index, np.zeros(len(pivot.index))
    for j, col_ev in enumerate(pivot.columns):
        ax.bar(years, pivot[col_ev], bottom=bottoms, label=col_ev, color=ev_cc
        bottoms += pivot[col_ev].values
    style_ax(ax, title=lbl, xlabel='Model Year', ylabel='Count')
    add_badge(ax, lbl, clr)
plt.tight_layout(pad=2.0); plt.show()

# Violin Plot: Range by CAFV
cafv_short = {'Clean Alternative Fuel Vehicle Eligible': 'CAFV Eligible', 'Not
```

```
fig, axes = plt.subplots(1, 3, figsize=(22, 7), facecolor='#0F1117')
for ax, dset, lbl, clr in zip(axes, datasets, labels, colors):
    sub = dset.copy(); sub['Clean Alternative Fuel Vehicle (CAFV) Eligibility'
    cats = sub['Clean Alternative Fuel Vehicle (CAFV) Eligibility'].value_cour
    sub = sub[sub['Clean Alternative Fuel Vehicle (CAFV) Eligibility'].isin(ca
    parts = ax.violinplot([sub[sub['Clean Alternative Fuel Vehicle (CAFV) Elig
    for pc, pc_clr in zip(parts['bodies'], ev_colors): pc.set_facecolor(pc_clr
    ax.set_xticks(range(1, len(cats) + 1)); ax.set_xticklabels(cats, rotation=
    style_ax(ax, title=lbl, xlabel='CAFV Status', ylabel='Electric Range'); ad
plt.tight_layout(pad=2.0); plt.show()
```



In [15]:
```
# Make x EV Type Heatmap
fig, axes = plt.subplots(1, 3, figsize=(22, 8), facecolor='#0F1117')
for ax, dset, lbl, clr in zip(axes, datasets, labels, colors):
    top_makes = dset['Make'].value_counts().head(12).index
    pivot = (dset[dset['Make'].isin(top_makes)].groupby(['Make', 'Electric Veh
    sns.heatmap(pivot, ax=ax, cmap='Blues', annot=True, fmt='d', annot_kws={'s
    ax.set_title(lbl, fontweight='bold', color=clr)
plt.tight_layout(pad=2.0); plt.show()

# Bubble Chart: Year vs Make
fig, axes = plt.subplots(1, 3, figsize=(22, 8), facecolor='#0F1117')
for ax, dset, lbl, clr in zip(axes, datasets, labels, colors):
    top_makes = dset['Make'].value_counts().head(10).index
    grp = (dset[dset['Make'].isin(top_makes)].groupby(['Model Year', 'Make']).
    sc = ax.scatter(grp['Model Year'], grp['Make'], s=grp['Count'] / grp['Cour
    style_ax(ax, title=lbl, xlabel='Model Year', ylabel='Make'); add_badge(ax,
plt.tight_layout(pad=2.0); plt.show()

print("\nEDA Notebook Generation Complete!")
```



EDA Notebook Generation Complete!